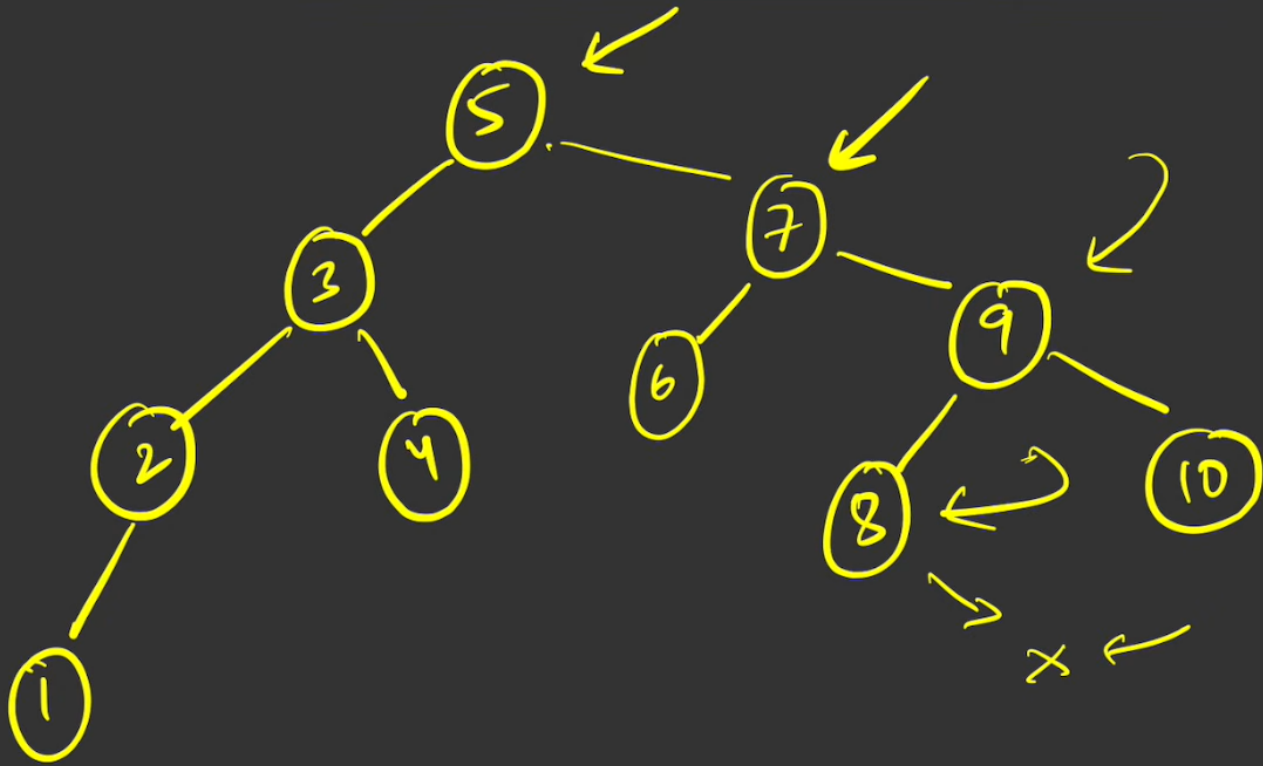


Inorder Successor in BST



val = 8

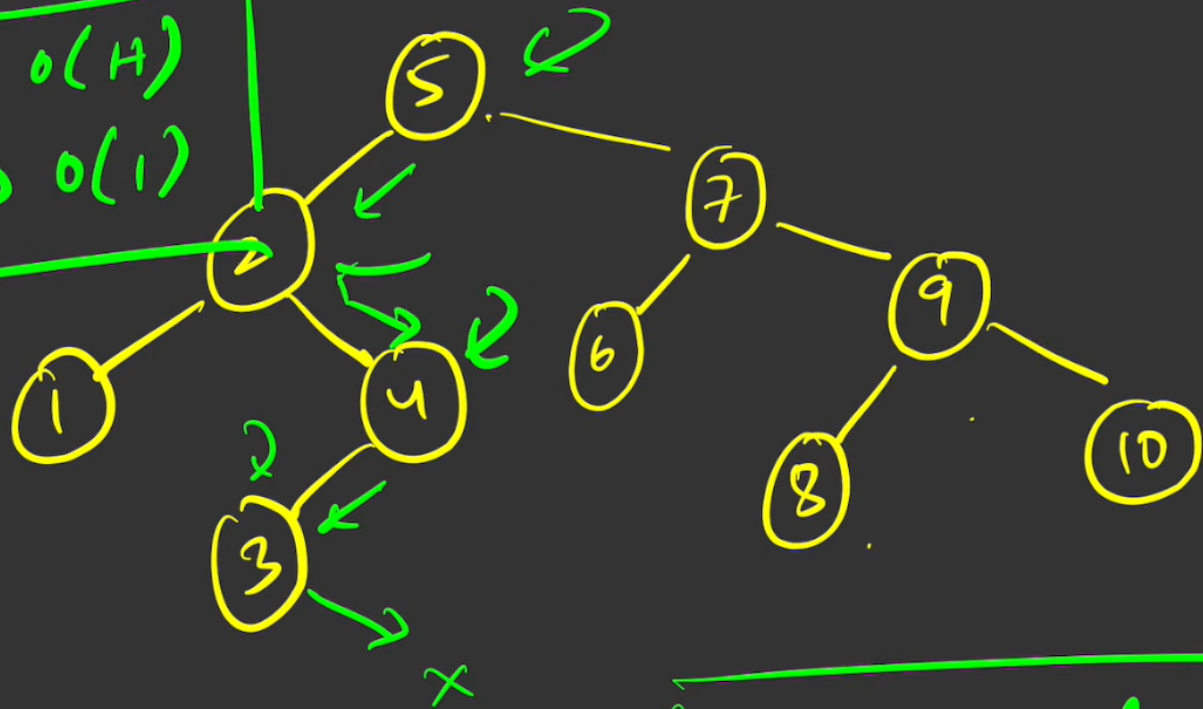
successor = ~~next~~ 9



Inorder Successor in BST

TC $\rightarrow O(H)$
SC $\rightarrow O(1)$

val = 3



Successor = next $\rightarrow$ 4



```
1  ▼ /**
2      * Definition for a binary tree node.
3      * struct TreeNode {
4      *     int val;
5      *     TreeNode *left;
6      *     TreeNode *right;
7      *     TreeNode(int x) : val(x), left(NULL), right(NULL) {}
8      * };
9      */
10 ▼ class Solution {
11     public:
12     ▼     TreeNode* inorderSuccessor(TreeNode* root, TreeNode* p) {
13         ▼         TreeNode* successor = NULL;
14
15         ▼         while (root != NULL) {
16
17             ▼             if (p->val >= root->val) {
18                 root = root->right;
19             ▼             } else {
20                 ▼                 successor = root;
21                 root = root->left;
22             }
23         }
24         ▼         return successor;
25     }
26 }
27 ;
```